Automatic Identification System Frequently Asked Questions

NOAA Office for Coastal Management

https://marinecadastre.gov/ais

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AIS Broadcast Points

Do you publish live or near real-time data?

Live data feeds are not available through this platform. The latest available data are hosted on our website. We do not restrict or embargo any data; data are published as soon as processing is complete.

Do you have data before 2009?

No, 2009 is the earliest year for which data are available.

Do you have AIS for my location?

Data are generally available for the U.S. Exclusive Economic Zone, major inland waterways, the Great Lakes, and Guam.

Coverage is currently unavailable for remote Pacific territories, foreign waters, or waters extending more than 40 to 50 miles from the coast. *Please see the Alaska-specific note below regarding historical coverage*.

What is your data source?

These data originate from the Nationwide Automatic Identification System (NAIS) operated by the U.S. Coast Guard. The NAIS comprises approximately 200 land-based receiving stations located near important navigation routes in the contiguous United States, Alaska, Hawaii, Guam, and parts of the Caribbean. This collection also includes data provided by the U.S. Army Corps of Engineers for inland waterways.

Do you have AIS data collected by satellite?

No. All data provided here originate from land-based receivers. While some commercial providers collect and sell satellite AIS data, and some federal agencies purchase these data for specific needs, those agencies are restricted from distributing satellite data to the public under the terms of their licensing agreement.

What format is used to publish the data?

- Prior to 2015: Data were published in File Geodatabase format.
- 2015 and onward: All broadcast points are provided in compressed CSV format.

For a year-by-year summary with additional technical details, please refer to the Point Data Summary table on the AIS home page.

How can I acquire an entire year of data?

To access daily nationwide files from 2015 onward, use the bulk download links on the AIS home page.

For requests under 2 GB, use the AccessAIS application to download data for a custom geographic area and time period.

Annual products for track and transit counts can be found in the download table on the AIS home page. These products are derived from original broadcast points and are often easier for casual users to work with because of their smaller file size and efficient data structure.

Why are data missing for Alaska in certain years?

Broadcast point data are available for Alaska through March 28, 2021. Track and transit count grids do not exist for this time period, and data gaps are present in the Northern and Arctic regions.

As of March 29, 2021, all Alaska broadcast point data were removed from this collection at the request of the Marine Exchange of Alaska, which owns most of the regional AIS receiving stations.

Why are some files considerably smaller than the average size?

Vessel traffic volume varies due to natural and human factors, as well as data-collection interruptions within the NAIS sensor network. Data received from the U.S. Coast Guard typically do not specify the cause or duration of sensor outages.

We use all records provided by the U.S. Coast Guard and employ quality-control techniques to document data gaps. Most gaps are brief—lasting from a few hours to a couple of days—and are usually limited to a small number of receiving stations.

Why hasn't the number of cargo vessels increased over time?

Several factors could influence this trend:

1. Improved reporting accuracy: The U.S. Coast Guard launched educational campaigns to help vessel owners report their vessel types more accurately.
2. Increased vessel size: The average size of cargo vessels has increased, allowing more cargo to be moved per ship.
3. Route changes: Some vessels may alter their routes to travel further offshore, outside the range of the NAIS receivers, to avoid certain emission control areas.

Are accidents or other events removed from the data?

No. Collisions, allisions, and other accidents are preserved in the data. Raw data from the National Marine Electronics Association (NMEA) are down-sampled to the nearest whole minute. The dataset comprises all vessels detected by land-based antennas, excluding certain law enforcement and military vessels exempt for national security reasons. Many smaller vessels are exempt from AIS carriage requirements due to their length or displacement.

How can I analyze data coverage?

To summarize broadcast point and track data, use an H3 or S2 tessellation to enumerate the magnitude or presence of data across a grid.

Why do some records appear far from shore, beyond normal radio range?

Signals detected beyond the expected range of land-based antennas often result from the anomalous propagation of AIS signals (e.g., tropospheric ducting). These data should not be viewed as erroneous or satellite-derived. Receiving stations at high elevations frequently detect signals from great distances, especially from commercial vessels with tall infrastructure and high-powered transmitters.

Why are data present over land?

GPS is the primary source for vessel positions. Positional errors can arise from natural, human, or systematic factors. Furthermore, vessels in narrow inland waterways or those transported by land may report coordinates that place them on shore.

Are Class B data included?

Yes. Class B data are included, and the class designation is provided for each record after 2017. Class B data are known to be present in the data starting in 2015, possibly earlier.

Where are the AIS receiver stations located?

The AIS Base Stations dataset and data service can be accessed via the Marine Cadastre Data Registry.

Broadcast Point Attributes

Is a data dictionary available?

Yes, a link to the latest data dictionary (2015–2025) is located at the bottom of the AIS web page. Data for 2009 to 2015 have a different structure and are normalized across three related tables (fig. 1).

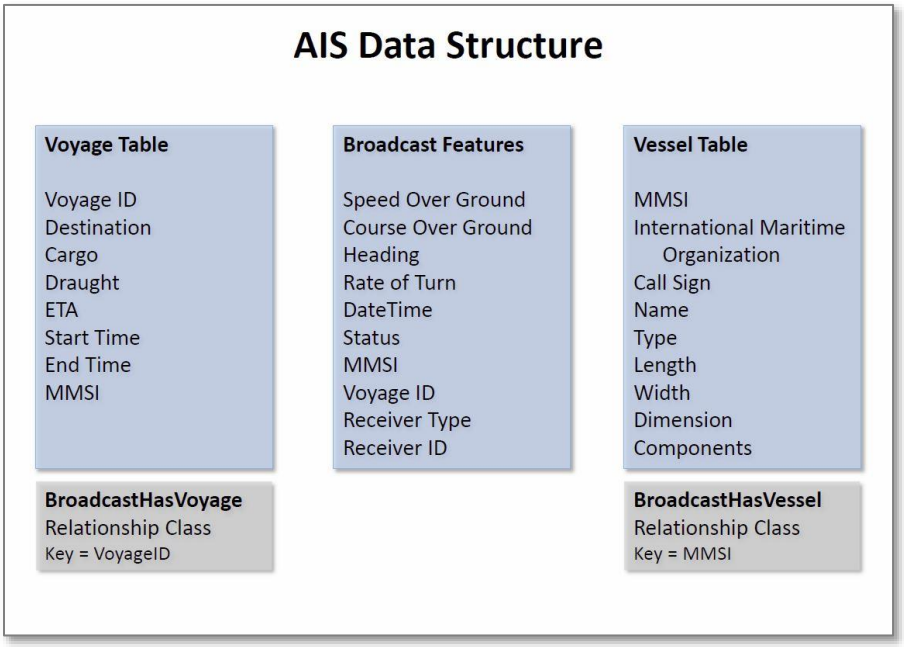


Figure 1. Simplified object model for 2009–2015 AIS broadcast points.

Have the vessel type codes been cleaned or modified?

Between 2015 and 2023, the U.S. Coast Guard's Authoritative Vessel Identification Service (AVIS) database was used to replace specific vessel-dependent values such as *vessel_type*, length, width, and draft. The original uncorrected *vessel_type* code from the raw NMEA record can still be found in the cargo field for that period.

Beginning in 2024, the AIS Vessel Identification Database (AVID) replaced AVIS. AVID is used primarily to populate null values or correct gross errors in vessel-dependent fields. For more details on vessel grouping, refer to NOAA's AIS Vessel Type and Group Codes (PDF).

What is the AIS Vessel Identification Database (AVID)?

AVID is used to verify and enhance vessel identity and characteristics. It integrates 23 sources including state, federal, and international authorities, as well as the final version of AVIS, into a fuzzy logic model. This model drives updates for null values and gross errors found in raw NMEA records.

AVID only modifies vessel-dependent values originally input by the mariner. Since its implementation in January 2024, AVID-based updates have corrected approximately 10–15 percent of records.

How do I interpret the vessel type and cargo codes?

The *vessel_type* codes are defined in the NMEA specification and documented in the Class A AIS Position Report (Messages 1,2,3) by the U.S. Coast Guard Navigation Center (NAVCEN). The USCG AIS Encoding Guide (PDF) and NOAA's AIS Vessel Type and Group Codes (PDF) are helpful resources.

For practical purposes, cargo codes are identical to vessel types. Discrepancies between *vessel_type* and cargo typically occur when corrections are imputed to the *vessel_type* value or the purpose of the vessel's voyage changes.

Why would a vessel name change over time?

Vessel names frequently change due to ownership transfers. Because *vessel_name* is a mariner input field, it is prone to human error or intentional modification. Sometimes, an AIS transponder (with its associated Maritime Mobile Service Identity [MMSI]) is transferred to a new owner; at other times, an owner may retain the transponder for use on an entirely different vessel.

When a name is uncertain or null in the raw NMEA data, the field is updated using the name associated with that Maritime Mobile Service Identity in AVID. For reliable tracking, use stable identifiers such as MMSI, IMO, or Call Sign.

Can I group data by vessel type to study trends?

Yes. Similar vessel types from native NMEA data and corrected vessel types can be combined into logical vessel groups (e.g., combining type 31 and type 1025 into "Towing"). A table of recommended vessel groups and their corresponding values is available in the Vessel Type and Group Codes (PDF).

Note: The "Other" vessel group contains a wide variety of specific vessel classifications.

What is the difference between vessel-type codes below and above 99?

- 0–99: Standardized NMEA vessel types
- 99–255: Reserved for future use
- Above 999: These are 4-digit AVIS service types. They are not part of the native NMEA stream.

From 2015 through 2017, 4-digit AVIS codes were used directly in the correction process. From 2018 through 2023, while AVIS was still used, these codes were mapped to the "best fit" NMEA vessel type (0–99) during processing.

The vessel type codes in the 1001–1025 range no longer exist. Is there a way to reclassify the 2-digit vessel type code to the 4-digit pattern?

Not exactly. While the 2-digit and 4-digit codes share similarities and near matches can be made for most analytical purposes, these codes do not have 1:1 direct mapping.

What does *vessel type 59* mean?

According to ITU Resolution 18, type 59 refers to “ships and aircraft of States not Parties to an armed conflict.”

Technical Guidance

How do I find historical data for a specific vessel?

Download the broadcast point data for your specific geographic and temporal area of interest, then query the dataset using the *mmsi, imo, call_sign*, or *vessel_name* field.

Can I unscramble encrypted MMSI values present in 2009–2014 data?

Yes, a brief user guide and standalone .exe tool for Windows are available in the MMSI Scrambler Tool ZIP file. Note: Technical support for this script is currently unavailable.

How do I find a vessel owner?

Vessel ownership is not recorded in this AIS dataset. However, the Maritime Mobile Service Identity can be used to combine this data with other public or licensed maritime databases to retrieve ownership information.

Are voyage-related data (e.g., Voyage ID, Destination) available?

These values were phased out in 2015 due to inherent inaccuracies. Destination values entered by mariners are often subjective and inconsistent. Vessel track datasets are available and defined by MMSI, start/stop times, and distance gaps.

How do I interpret Course Over Ground (COG) values?

Course Over Ground values are provided by the vessel's GPS unit. Errors can occur due to poorly configured hardware, phase wrapping, or non-north settings.

- Pre-2015 data: If a COG value is negative, calculate a new Course Over Ground using a pair of latitude and longitude coordinates.
- Sentinel values: A value of 360.0 indicates that COG data are unavailable and should be ignored.

Why do some Speed Over Ground (SOG) values in the pre-2015 data appear incorrect?

Speed Over Ground values are subject to the same GPS-related errors as COG values.

- Pre-2015 data: If a SOG value is negative, calculate a new Speed Over Ground using a pair of latitude and longitude coordinates.
- Sentinel values: A value of 102.3 indicates that SOG data are not available.
- Outliers: Extreme outliers are occasionally caused by AIS transponders mounted on aircraft rather than vessels.

Why do the time stamp and local time of an observed vessel differ?

All time stamps in the NMEA record use Coordinated Universal Time (UTC). Local time will differ based on time zone offsets and daylight saving time adjustments.

How do "Draught" (2009–2014) and "Draft" (2015–Present) compare?

- 2009–2014 (Draught): Reported in decimeters (1/10th of a meter). A value of 255 represents 25.5 meters or greater.
- 2015–Present (Draft): The American spelling was adopted. Units are reported in meters with 1/10-meter resolution.

Be aware that carriage requirements have changed; modern entries typically report the maximum present static draft.

Are parameters like tonnage, horsepower, or fuel type included?

No, these parameters are not part of the standard AIS broadcast. This information can be obtained from third-party maritime logistics vendors by using the Maritime Mobile Service Identifier as a relational key.

What does a value of "511" in the heading field represent?

In pre-2015 data, 511 was the default sentinel value that indicated heading data were not available.

AIS Track and Transit Counts

What is the cell size of the density grids, and what do the values represent?

The cell size for the 2011/2013 kernel density grids and post-2015 transit count grids is 100 meters.

- Kernel density: These values represent the count of features per unit area, typically visualized as a heat map.
- Transit counts: These values represent the actual number of unique vessel tracks passing through a specified cell.

While kernel density analyses generally use point data as an input, the Transit Count product is derived from the count of unique vessel tracks intersecting a grid cell.

How can I get an estimated time of arrival for a vessel?

Query the end time in the Vessel Track product for the line feature nearest to the destination. For a more precise estimate, query the *base_date_time* value for the broadcast point nearest to the destination.

Why is there a small irregularity in track and transit count data, running north–south in the Gulf of America?

The track data are constructed on a per ocean basin pattern and then merged into a national dataset. This process can result in a minor gap or overlaps west of Florida, where the Gulf of America and Atlantic data subsets meet.

Why are track data absent for Alaska for some years?

Access to broadcast point data for Alaska has been intermittent. Therefore, derived data products, such as tracks and transit counts, have not been generated.

How do I work with the GeoPackage data format?

First, unzip the compressed file to extract the .gpkg file. The GeoPackage data object is an SQLite container that holds all the geometry and attribute data. This file can be opened directly by most modern desktop GIS platforms and accessed via Python libraries.

AccessAIS Provisioning System

What types of AIS data are available from the AccessAIS tool?

AccessAIS is a data provisioning system for broadcast point data. Track and transit count data are not distributed by this application. However, transit count and other layers are included in the map display

to help users orient themselves during the ordering process. New broadcast point data are added to the tool quarterly; transit count display layers are updated annually.

My order says "Download has expired." What should I do?

The CSV download link expires after 14 days or once accessed five times, whichever occurs first. If your link has expired, please submit a new order. Expired orders cannot be recovered or resubmitted.

Why is there a size limit for the download?

Size limits maintain system performance, control costs, and serve the broadest possible audience. For larger data requests, please use the *Bulk Download* page.

If records exist back to 2009, why does AccessAIS only host the last five years of data?

Design and cost constraints during tool development determined the maximum storage capacity. That limit resulted in a five-year rolling order window for available data. Data older than five years remain available for download from the *Bulk Download* page.

Does AccessAIS source data originate from a live U.S. Coast Guard feed?

No, these data originate from quarterly historical data requests to the U.S. Coast Guard. The raw data is then down-sampled and filtered to generate a derived data product for public use. While data from AccessAIS and the Bulk Download site differ slightly in format and structure, a 2026 upgrade is scheduled to eliminate these differences.

Are data for inland areas included in AccessAIS?

Yes, data for inland areas are included for locations where the U.S. Coast Guard and the U.S. Army Corps of Engineers maintain antennas. The presence or absence of data shown in the vessel track or vessel transit count product is not an absolute indicator of broadcast point data availability. The broadcast point data are the superset from which vessel track and transit count data are built. In some cases, broadcast point coverage is sparse, and the track and transit count data products are unpopulated.

Are data available for Alaska?

Coverage in Alaska is inconsistent, both temporally and geographically. Good coverage existed from 2015 to 2017 (especially in northern Alaska), but data are not available after March 29, 2021. Efforts to backfill these data gaps are ongoing.

Are data available for Guam and the Commonwealth of the Northern Mariana Islands?

These data are not within AccessAIS at this time. However, broadcast point data for this region can be acquired via the Bulk Download table and the annual Guam ZIP file folder. Track and transit count data products for this area were not generated until 2025.

How often are new data added to the AccessAIS application?

New broadcast point data are added every 90 days; however, delays may occur due to government shutdowns or other unforeseen events. The total lag time from data collection to delivery is approximately 145 to 165 days.

Can I obtain a URL for my area of interest to share with others?

Once your order has been processed, you will receive a shareable URL to download data. The download can be accessed up to five times within a 14-day window. Download errors can often be resolved by clearing the browser cache.

Appropriate Use

Can I use these data for regulatory or enforcement purposes?

These data are intended for coastal and ocean planning purposes only.

Can I use and redistribute these data on other websites?

Yes, you may release derived products built from these data. Please cite your data source to promote transparency.

Who owns the copyright to these data? Can I use it to generate a commercial product?

According to the U.S. Coast Guard Navigation Center:

> *These data are derived from the U.S. Coast Guard NAIS subject to the conditions of data sharing category "Level C (Historical Data)" and within the framework of Commandant Instruction 5230.80, Sharing of Information Collected by the Coast Guard Nationwide Automatic Identification System. The historical data are generally considered public domain.*
>
> *All provided data may not be used for purposes other than those intended for the disclosure. Foreign governments, Federal, State, local and tribal government agencies, and non-governmental entities shall not retransmit or redistribute AIS information (real-time or stored) in any form other than those intended for the disclosure as approved, and shall not charge a fee for its usage.*

Does NOAA have a terms and condition statement for these data?

> *As required by 17 U.S.C. § 403, third parties producing copyrighted works consisting predominantly of the material produced by U.S. government agencies must provide notice with such work(s) identifying the U.S. Government material incorporated and stating that such material is not subject to copyright protection within the United States. The information on*

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Citations and Points of Contact

How do I cite Marine Cadastre data?

NOAA Office for Coastal Management. ([*Year*]). *[Title of Dataset]*. Marine Cadastre. https://marinecadastre.gov.

Who should I list as the point of contact when citing the Marine Cadastre AIS data?

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